Event Segmentation Quality Analysis

P&S Arch4Health — Understanding Noise in Genome Sequencers

Datasets:
D1: SARS-CoV-2 (R9.4)
D6: E. coli (R10.4)
60 reads analyzed, 5 segmentation algorithms

Categories:
A: Nanopore-specific tool (Scrappie)
B: Nanopore-specific algorithms (ONT t-stat, HMM)
C: General changepoint baselines (PELT, BinSeg)

SAFARI Research Group — ETH Zürich
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Quality Metrics Explained

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Segmentation Quality Metrics:

1. SNR (Signal-to-Noise Ratio) [dB]
   = 10 * log10( variance_between_segments / variance_within_segments )
   Higher = better. Measures how distinct the detected events are.
   Think of it as: are the segment means spread out (signal) while
   within each segment the values are consistent (low noise)?

2. Variance Reduction R^2
   = 1 - (weighted_intra_variance / global_variance)
   Range [0, 1]. Fraction of total signal variance "explained" by
   the segmentation. R^2=1 means the segmentation perfectly captures
   all signal structure.

3. Normalized Intra-Segment Variance
   = intra_variance / global_variance
   Lower = better. Measures residual noise within segments.

4. Event Length Statistics
   Mean, median, CV of event durations (in samples).
   For R9.4 at 4kHz: typical event = 5-15 samples (~1-4 ms per nucleotide).

5. Number of Events
   More events != better. Over-segmentation splits real events.
   Under-segmentation merges distinct levels.

Algorithm Categories:
  A: Nanopore-specific compiled tools (Scrappie, RawHash2)
  B: Nanopore-specific algorithms in Python (ONT t-stat, HMM)
  C: General-purpose changepoint detection baselines (PELT, BinSeg)
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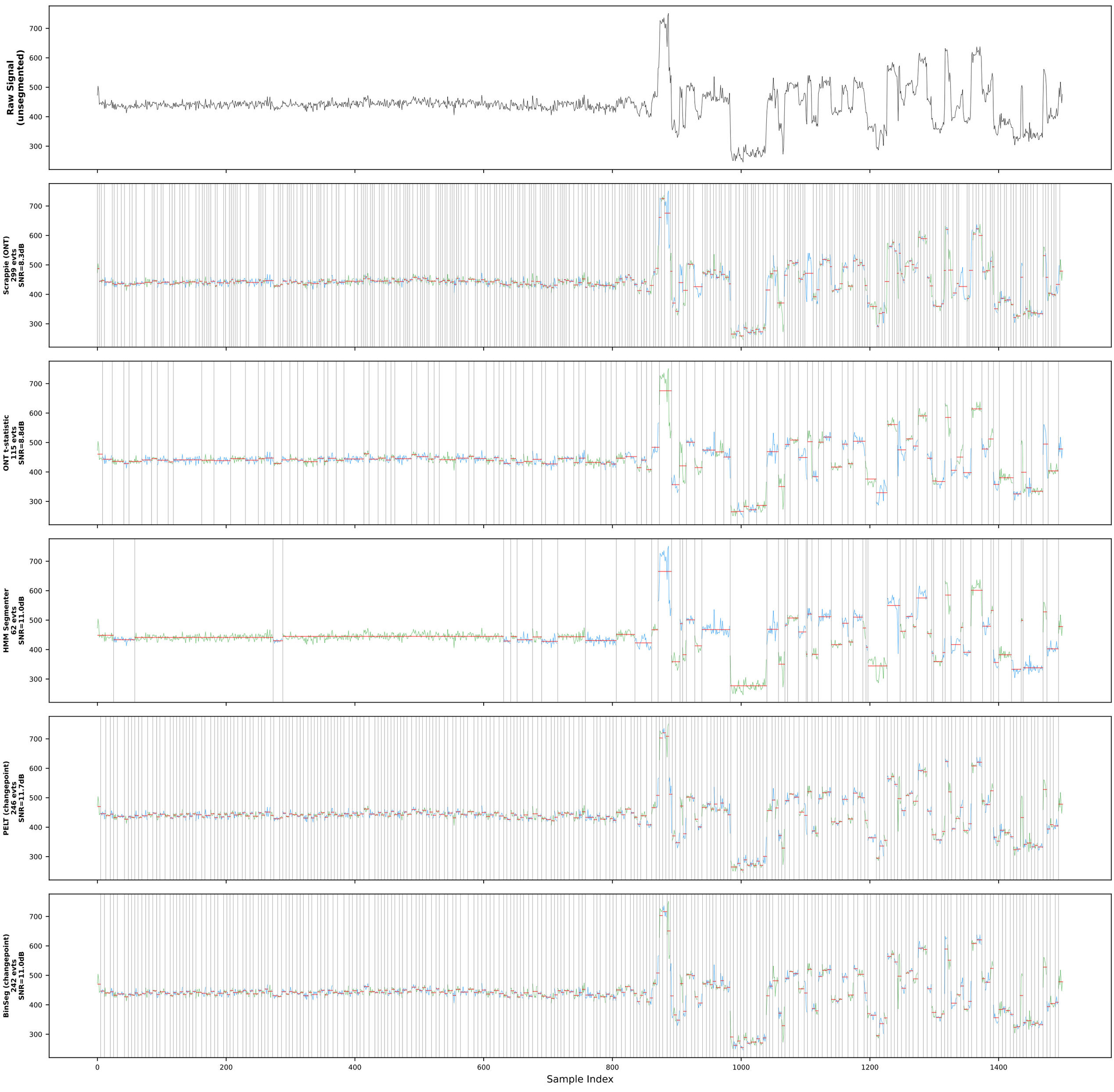
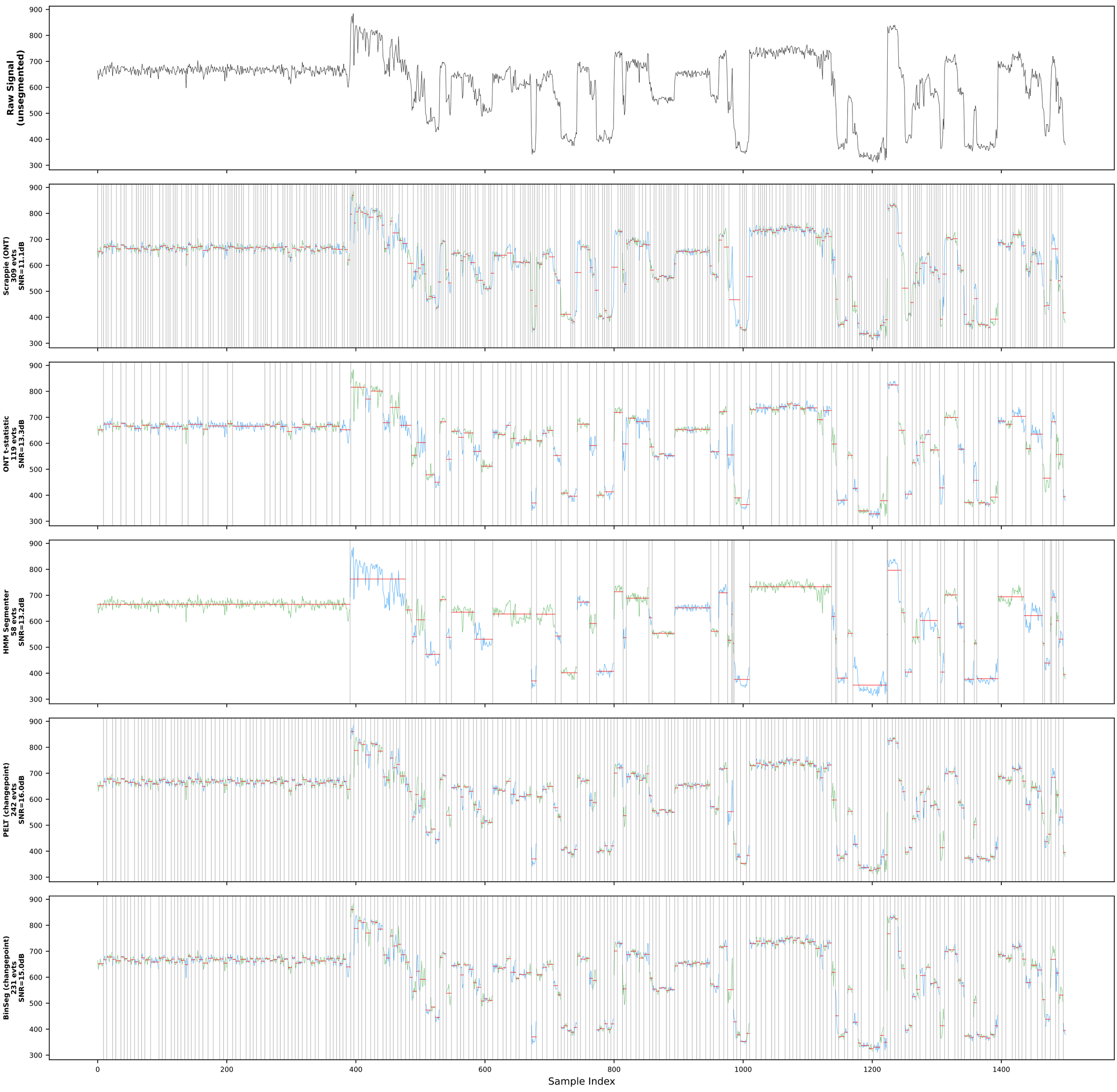
Segmentation Quality Summary — D1: SARS-CoV-2 (R9.4)

Algorithm	Category	Avg Events	SNR (dB)	R^2	Norm Intra	Mean Evt Len	CV Evt Len	Avg Time (s)
Scrappie (ONT)	A: Nanopore tool	770	7.08	0.7515	0.1822	5.4	1.97	0.1161
ONT t-statistic	B: Nanopore algorithm	351	8.64	0.8614	0.1202	11.9	0.39	0.1142
HMM Segmenter	B: Nanopore algorithm	307	10.38	0.8911	0.2134	13.7	1.56	0.3008
PELT (changepoint)	C: General baseline	682	11.59	0.9179	0.0942	6.1	0.23	2.4077
BinSeg (changepoint)	C: General baseline	661	10.93	0.9089	0.1044	6.3	0.24	5.4513

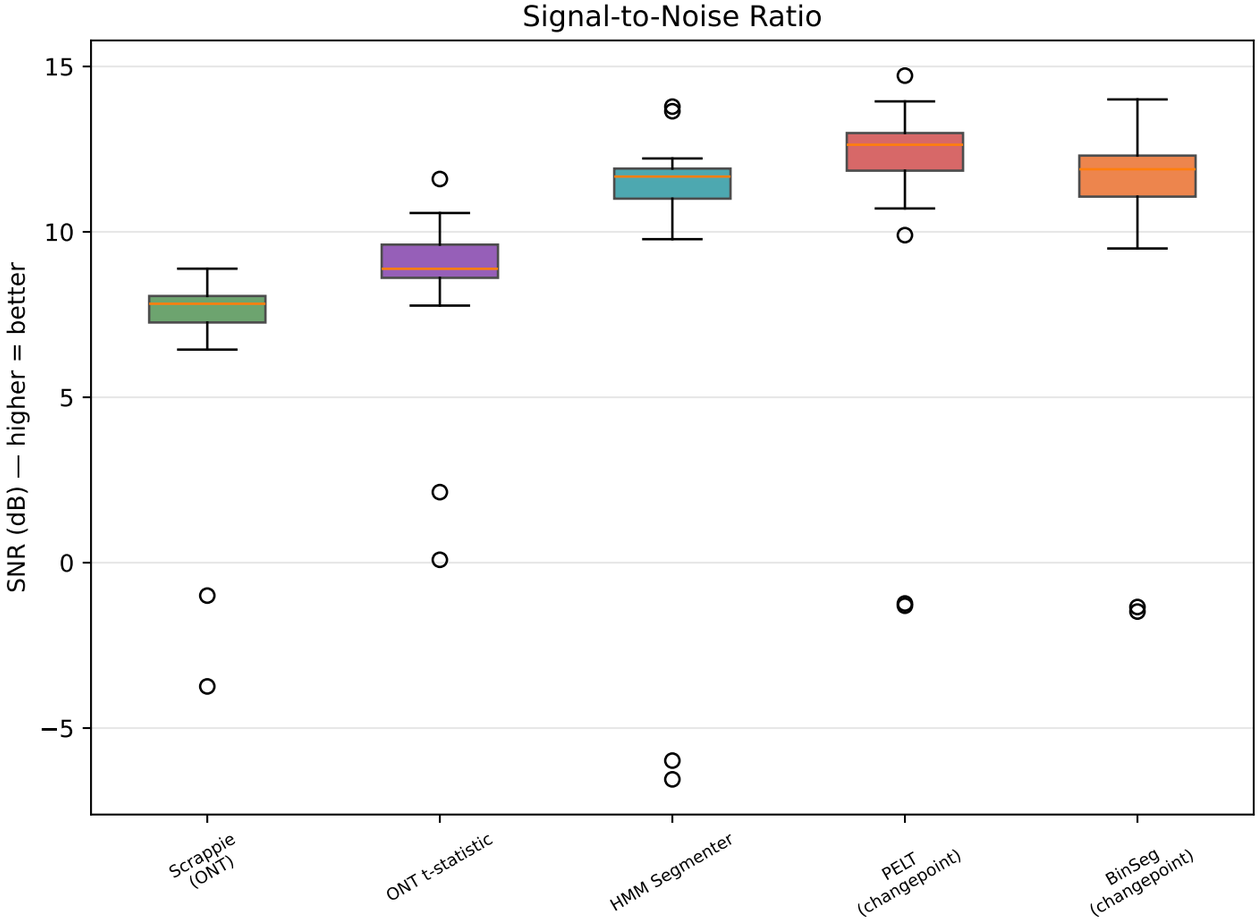
Segmentation Quality Summary — D6: E. coli (R10.4)

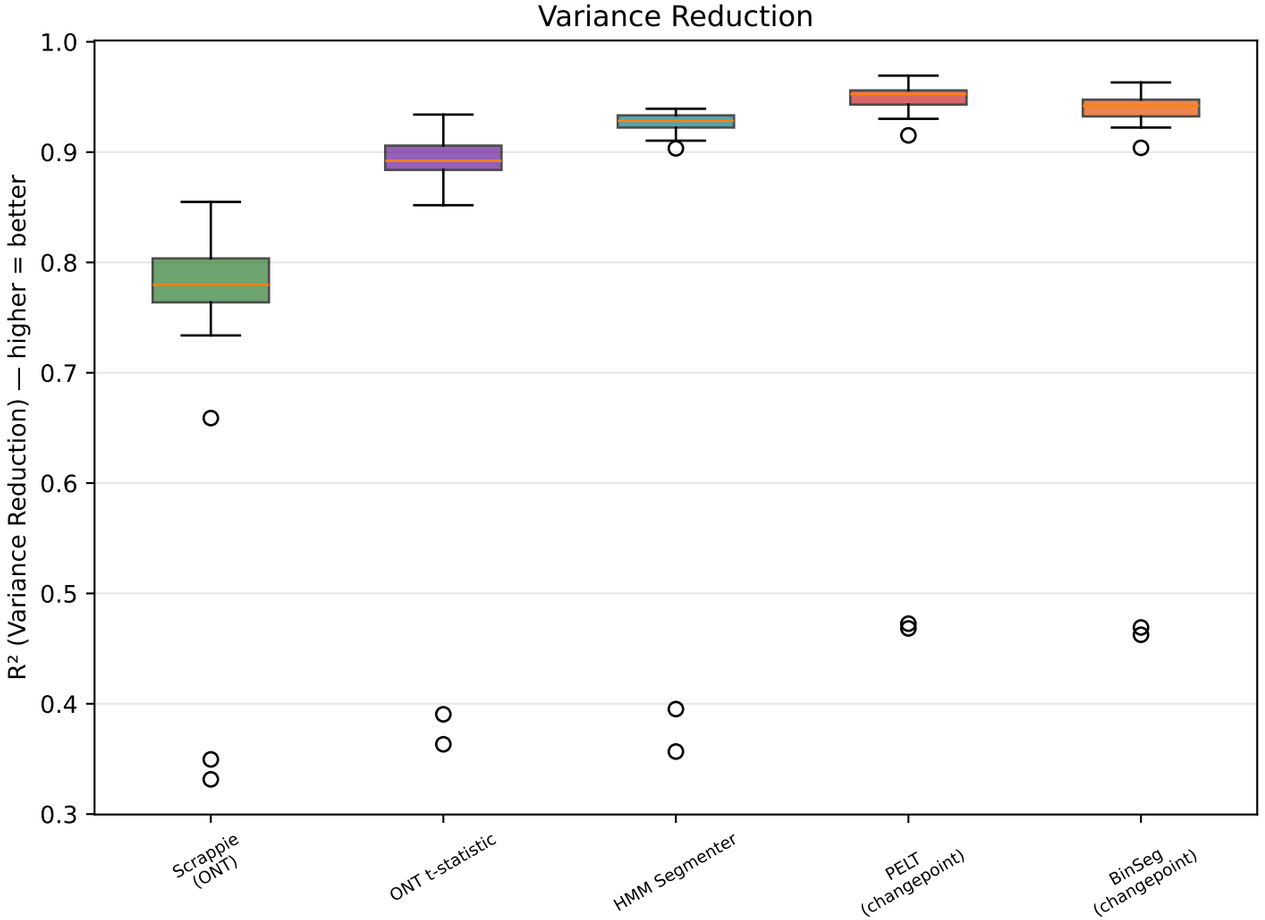
Algorithm	Category	Avg Events	SNR (dB)	R^2	Norm Intra	Mean Evt Len	CV Evt Len	Avg Time (s)
Scrappie (ONT)	A: Nanopore tool	5882	11.38	0.8934	0.0712	5.3	1.04	0.0823
ONT t-statistic	B: Nanopore algorithm	2372	14.07	0.9625	0.0395	12.9	0.42	0.8390
HMM Segmenter	B: Nanopore algorithm	1508	13.19	0.9599	0.0352	20.1	1.60	2.1726
PELT (changepoint)	C: General baseline	5019	16.75	0.9797	0.0218	6.1	0.23	161.5230
BinSeg (changepoint)	C: General baseline	4839	15.69	0.9743	0.0276	6.3	0.25	83.2571

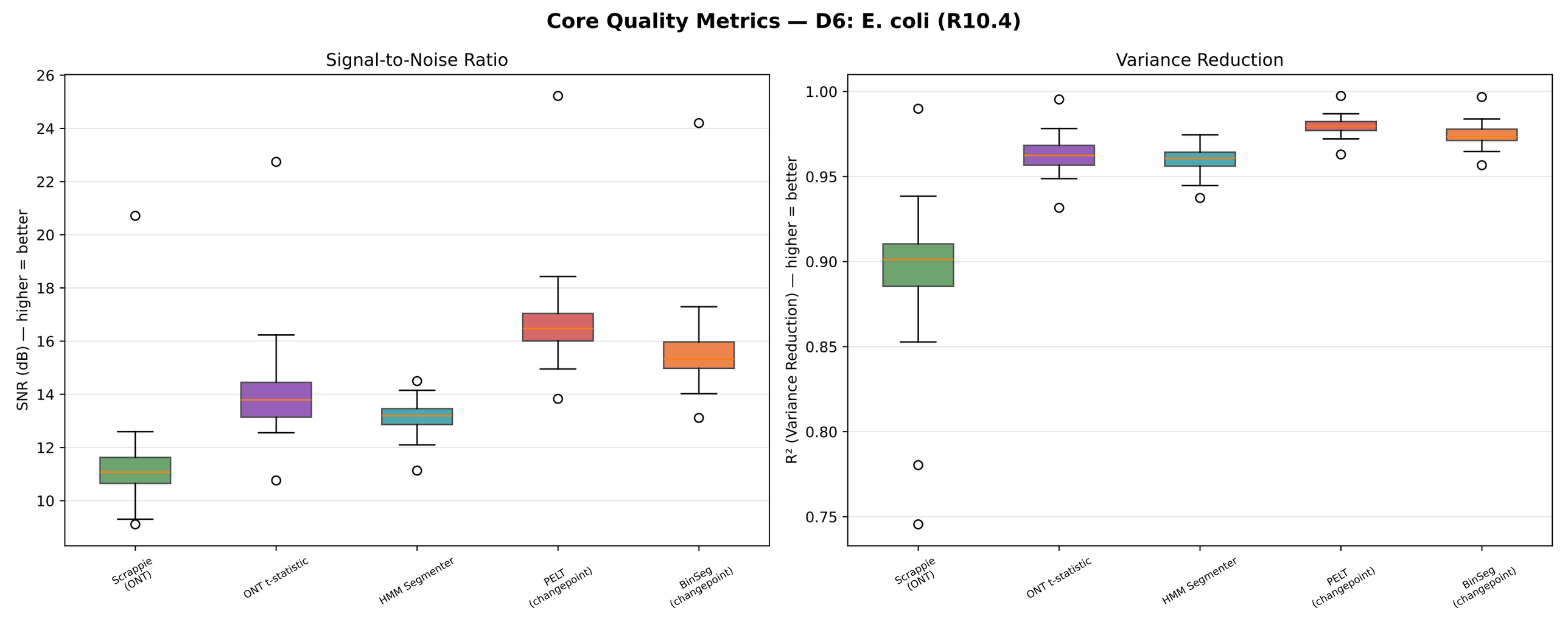

Event Segmentation Visualization — D1: SARS-CoV-2 (R9.4)
(First Read, First 1500 Samples)
Raw Signal
(unsegmented)
Scrappie (ONT)
299 evts
SNR=8.3dB
ONT t-statistic
115 evts
SNR=8.8dB
HMM Segmenter
62 evts
SNR=11.0dB
PELT (changepoint)
246 evts
SNR=11.7dB
BinSeg (changepoint)
242 evts
SNR=11.0dB
700
600
500
400
300
0
200
400
600
800
1000
1200
1400
Sample Index


Event Segmentation Visualization — D6: E. coli (R10.4)
(First Read, First 1500 Samples)
Raw Signal
(unsegmented)
Scrappie (ONT)
309 evts
SNR=11.1dB
ONT t-statistic
119 evts
SNR=13.3dB
HMM Segmenter
58 evts
SNR=13.2dB
PELT (changepoint)
242 evts
SNR=16.0dB
BinSeg (changepoint)
231 evts
SNR=15.0dB
Sample Index

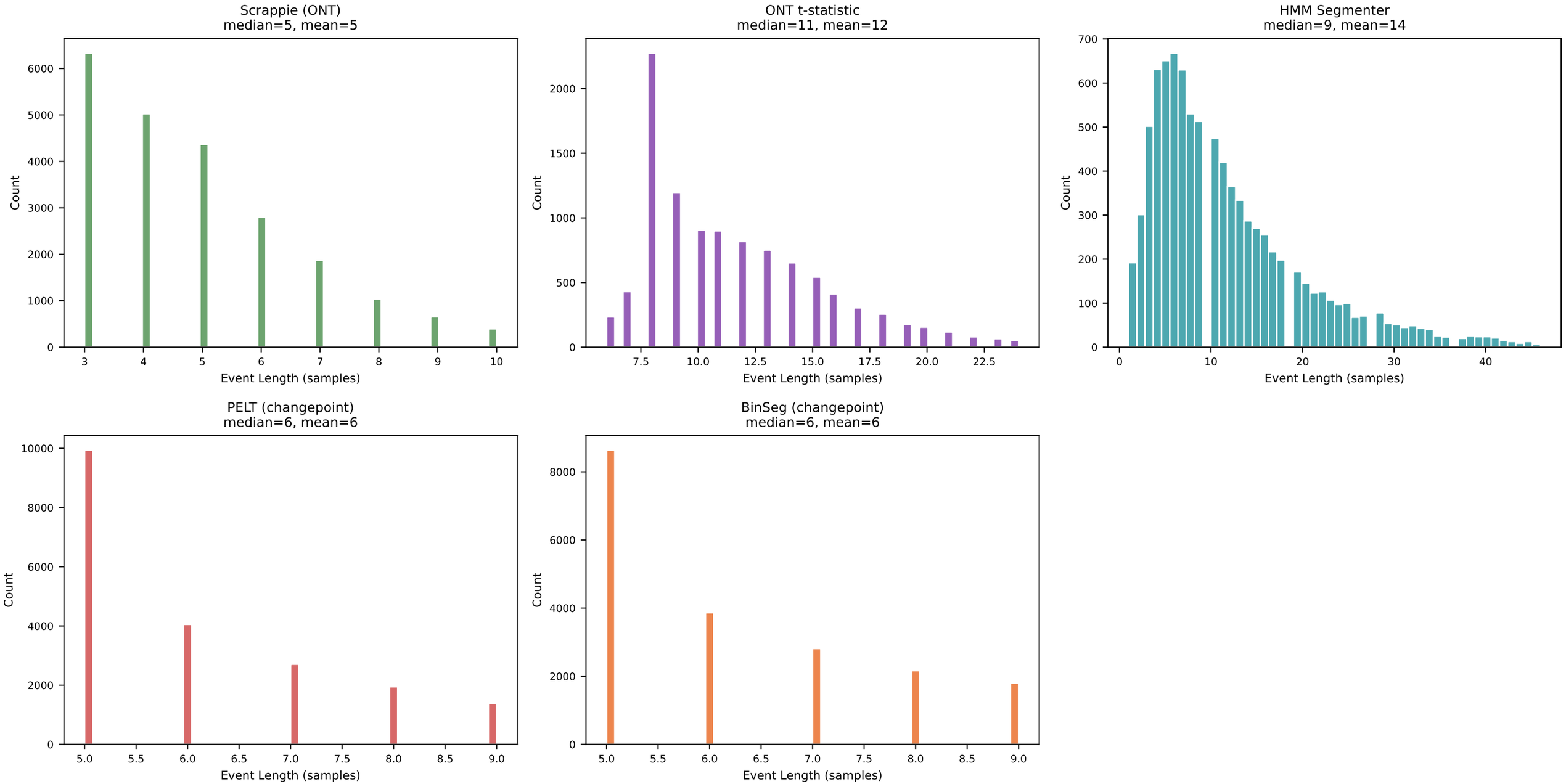
Core Quality Metrics — D1: SARS-CoV-2 (R9.4)



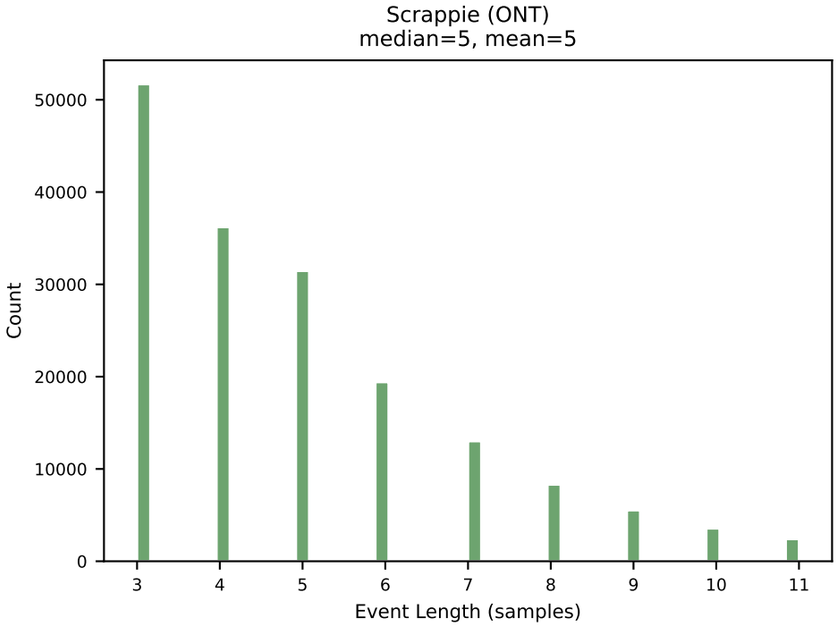


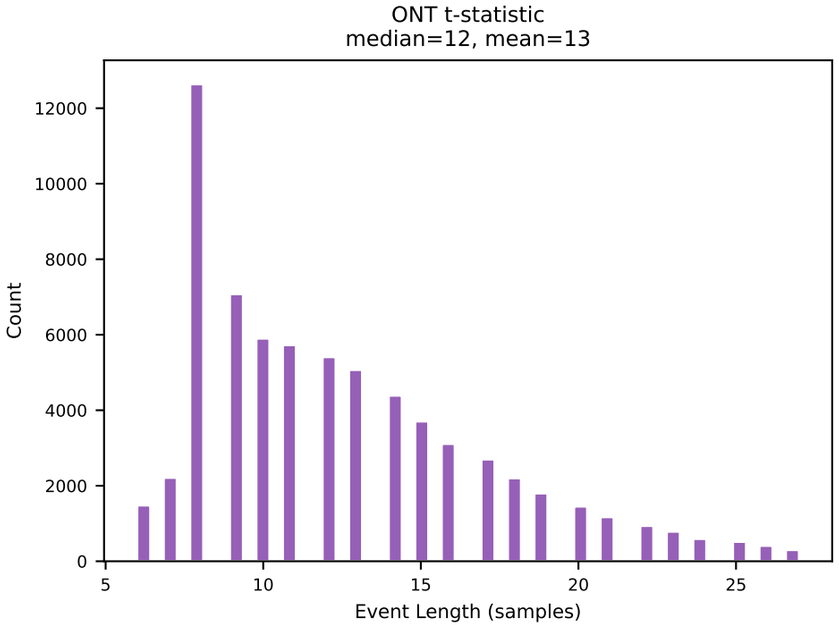

Core Quality Metrics — D6: E. coli (R10.4)
Signal-to-Noise Ratio
SNR (dB) — higher = better
26
24
22
20
18
16
14
12
10
Scrappie (ONT)
ONT t-statistic
HMM Segmenter
PELT (changepoint)
BinSeg (changepoint)
Variance Reduction
R² (Variance Reduction) — higher = better
1.00
0.95
0.90
0.85
0.80
0.75
Scrappie (ONT)
ONT t-statistic
HMM Segmenter
PELT (changepoint)
BinSeg (changepoint)

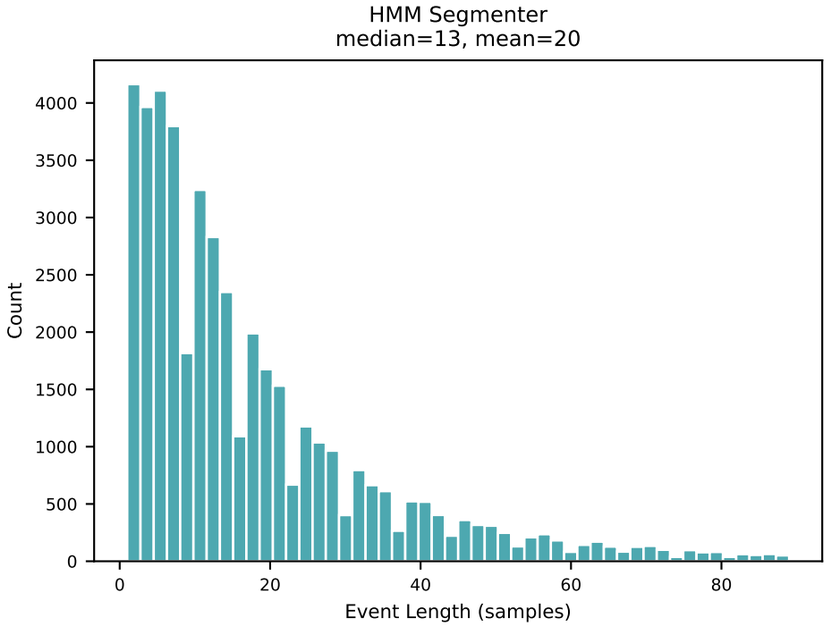
Event Length Distribution — D1: SARS-CoV-2 (R9.4)

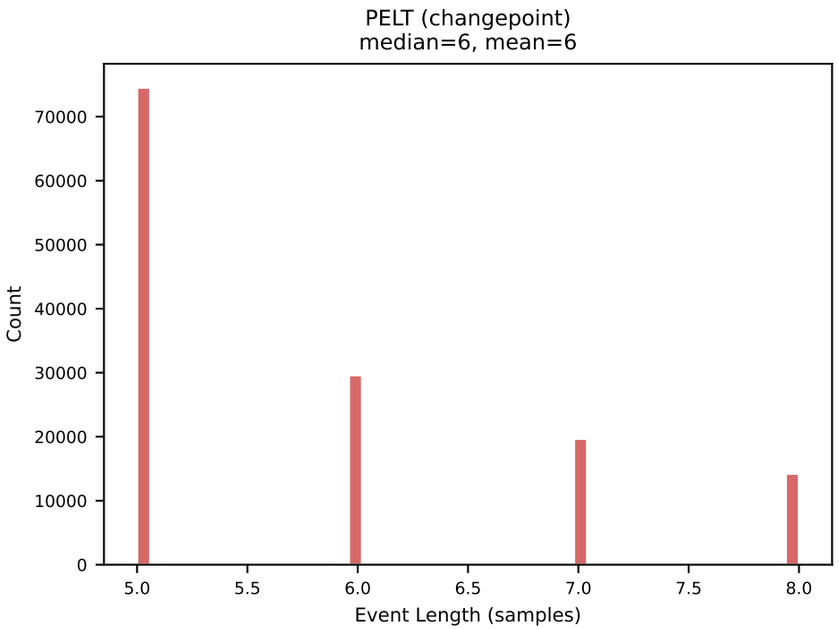


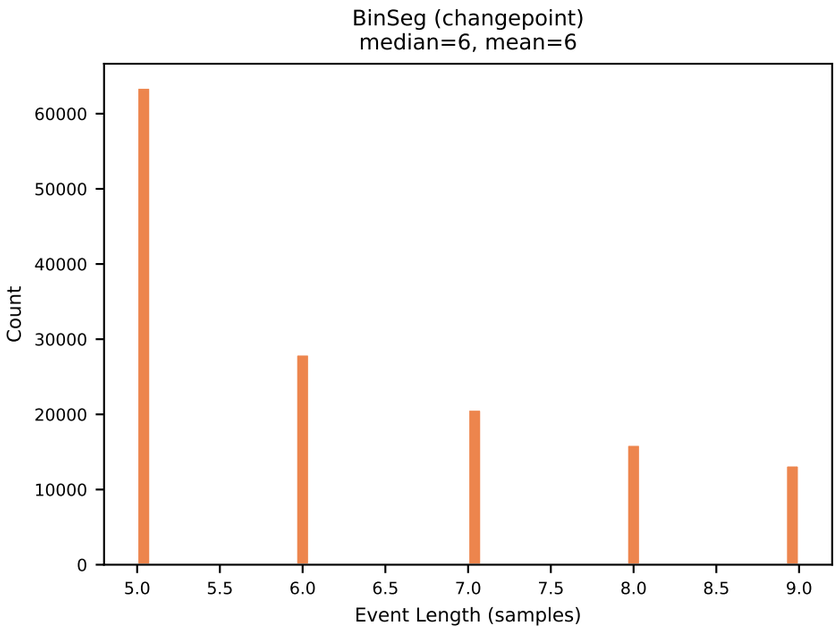
Event Length Distribution — D6: E. coli (R10.4)

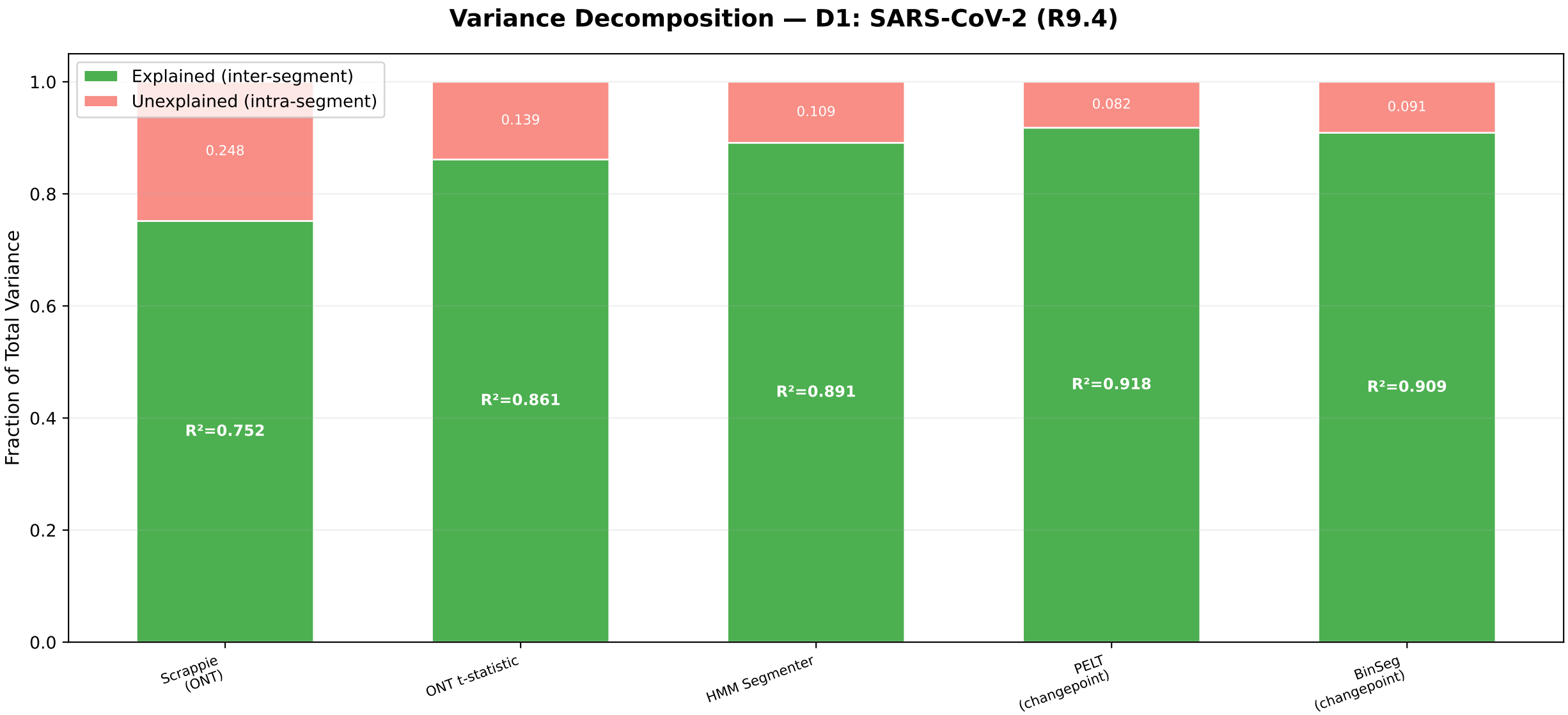


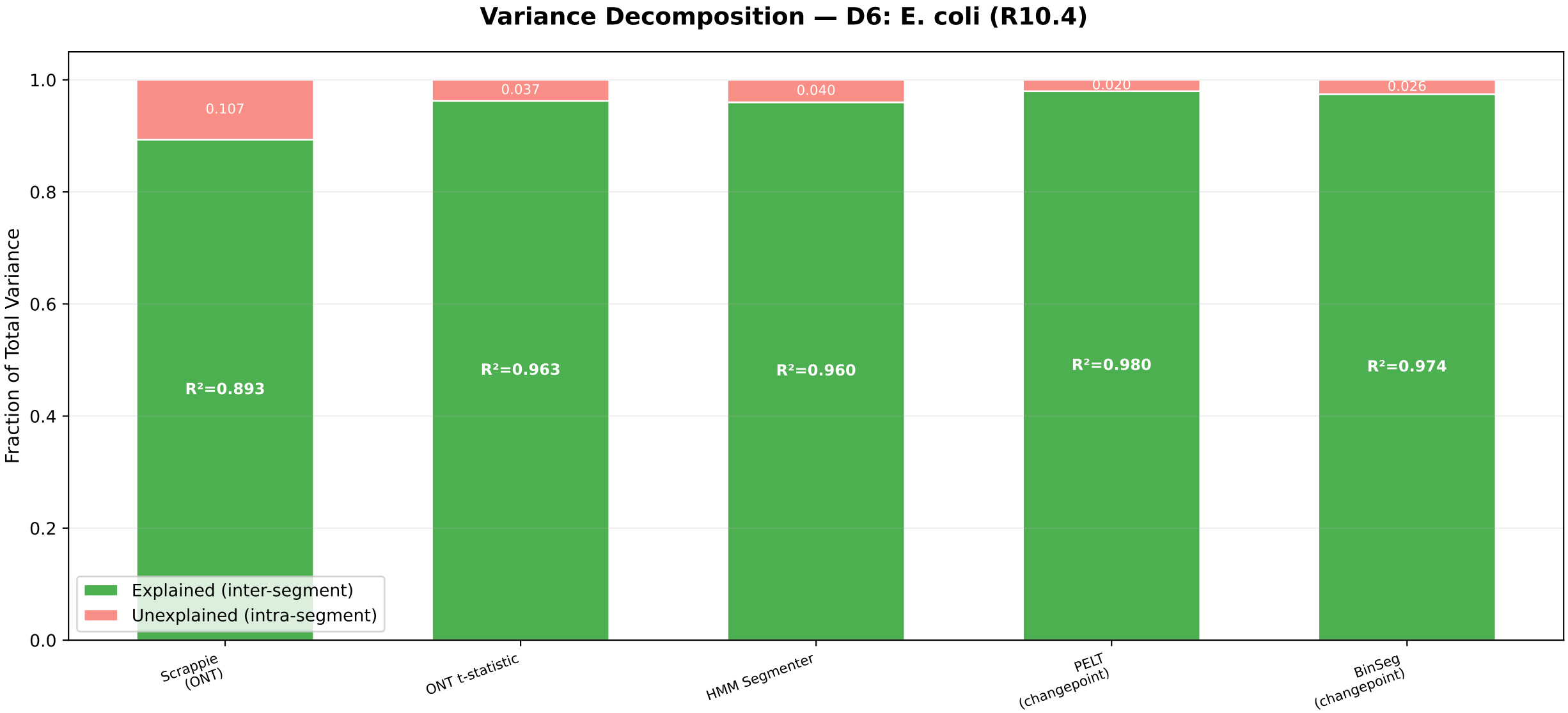




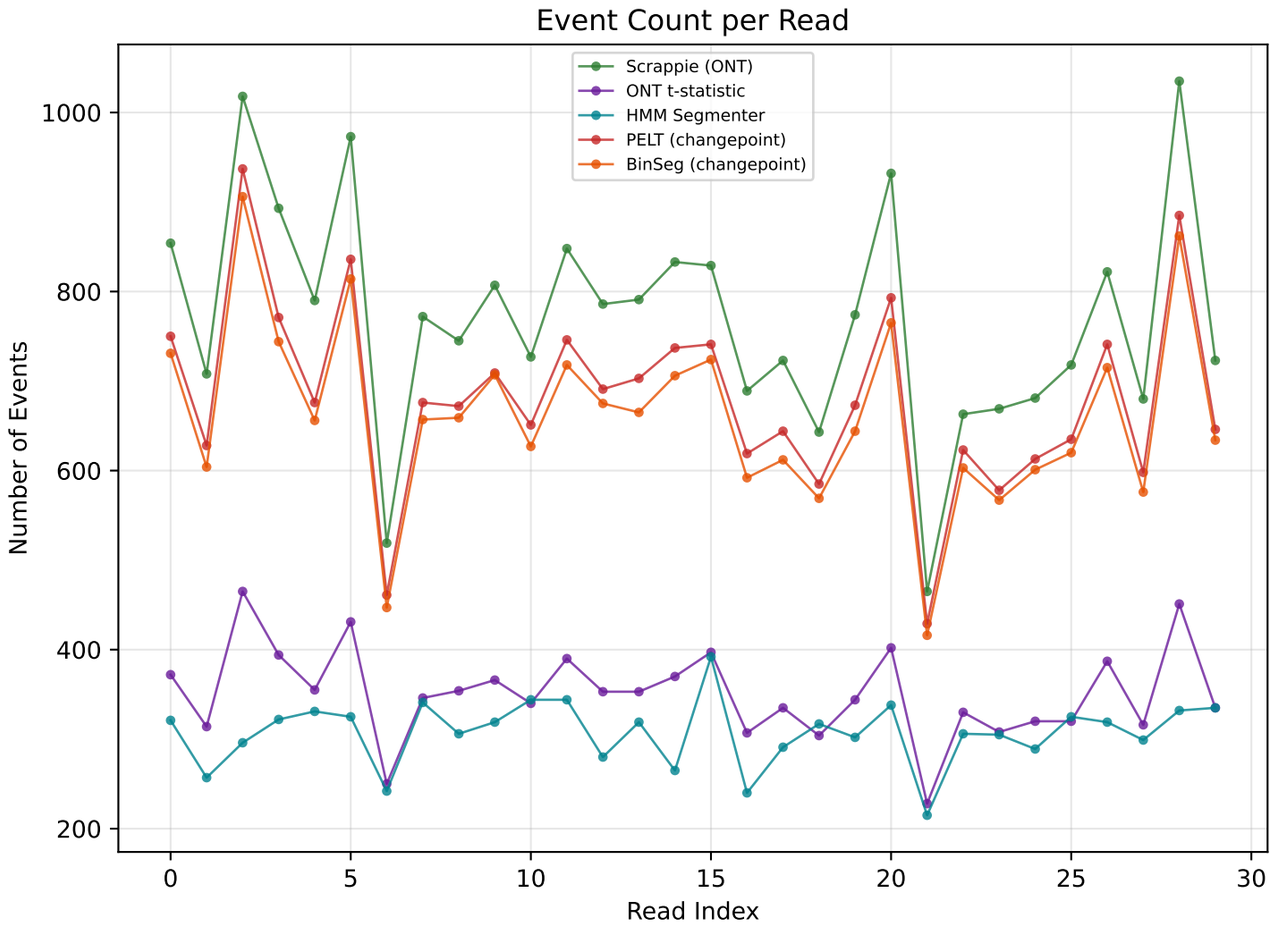


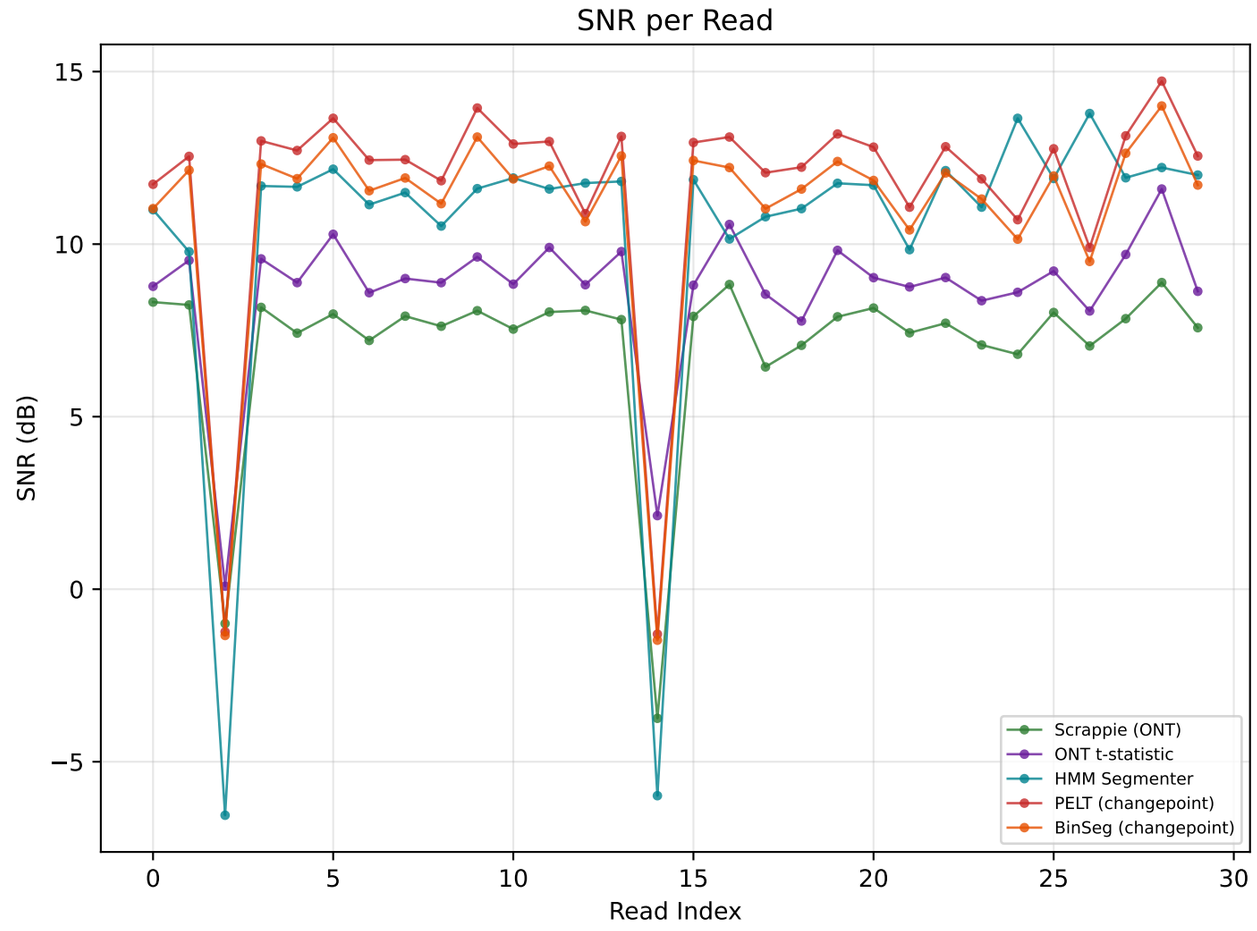



Variance Decomposition — D1: SARS-CoV-2 (R9.4)
Explained (inter-segment)
Unexplained (intra-segment)
Fraction of Total Variance
1.0
0.8
0.6
0.4
0.2
0.0
0.248
R²=0.752
0.139
R²=0.861
0.109
R²=0.891
0.082
R²=0.918
0.091
R²=0.909
Scrappie (ONT)
ONT t-statistic
HMM Segmenter
PELT (changepoint)
BinSeg (changepoint)

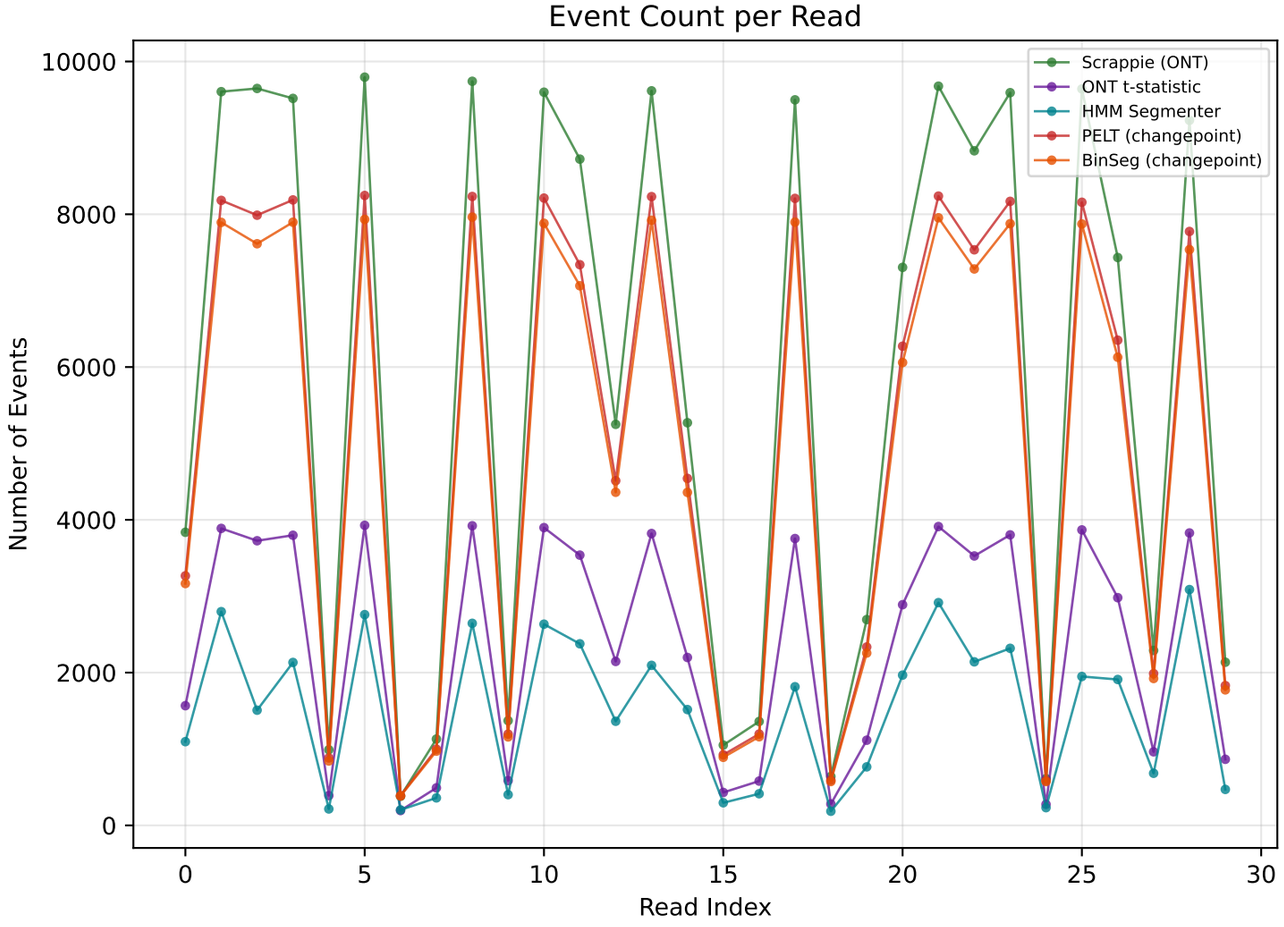

Variance Decomposition — D6: E. coli (R10.4)
Fraction of Total Variance
1.0
0.8
0.6
0.4
0.2
0.0
0.107
0.037
0.040
0.020
0.026
R²=0.893
R²=0.963
R²=0.960
R²=0.980
R²=0.974
Explained (inter-segment)
Unexplained (intra-segment)
Scrappie (ONT)
ONT t-statistic
HMM Segmenter
PELT (changepoint)
BinSeg (changepoint)

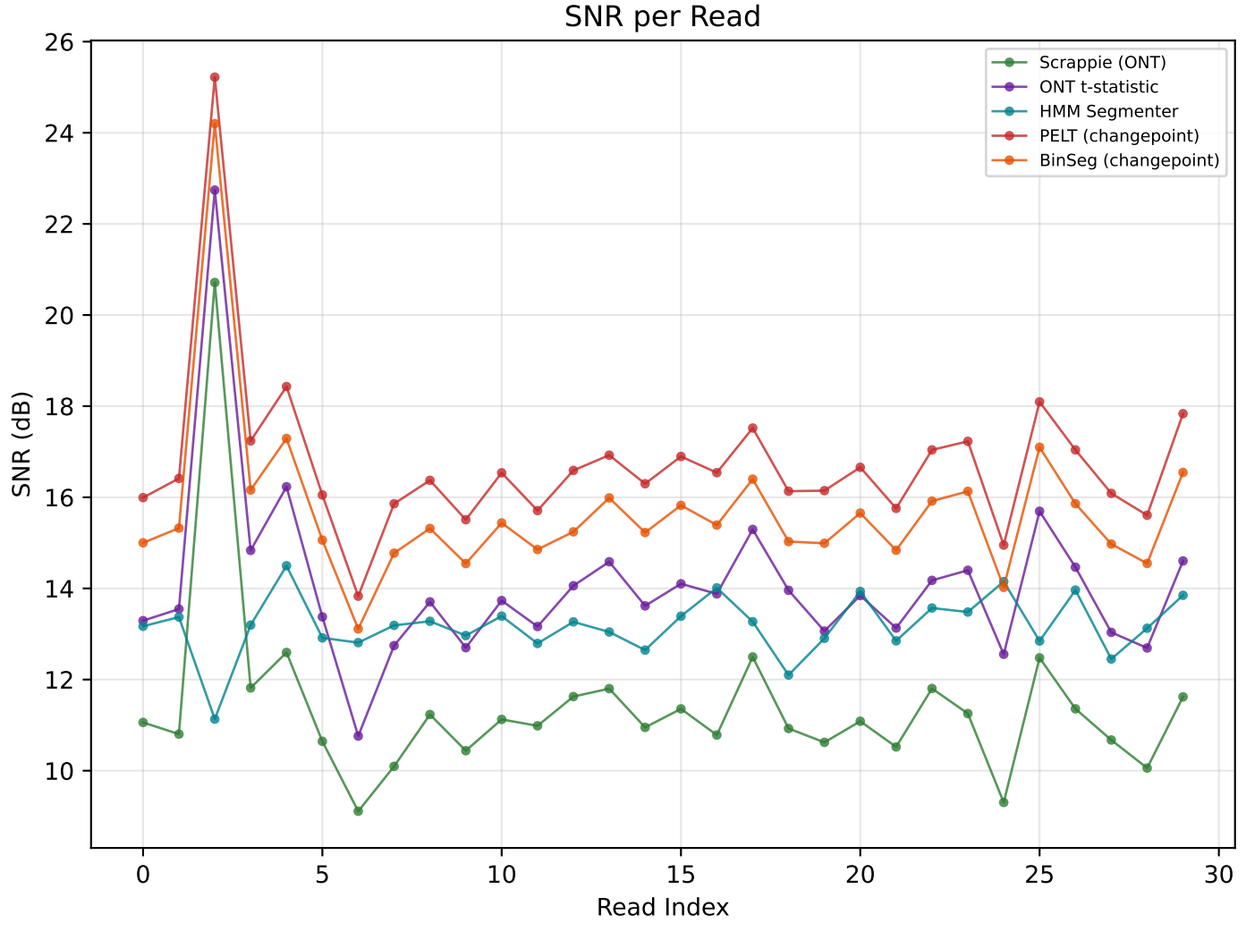
Segmentation Consistency Across Reads — D1: SARS-CoV-2 (R9.4)

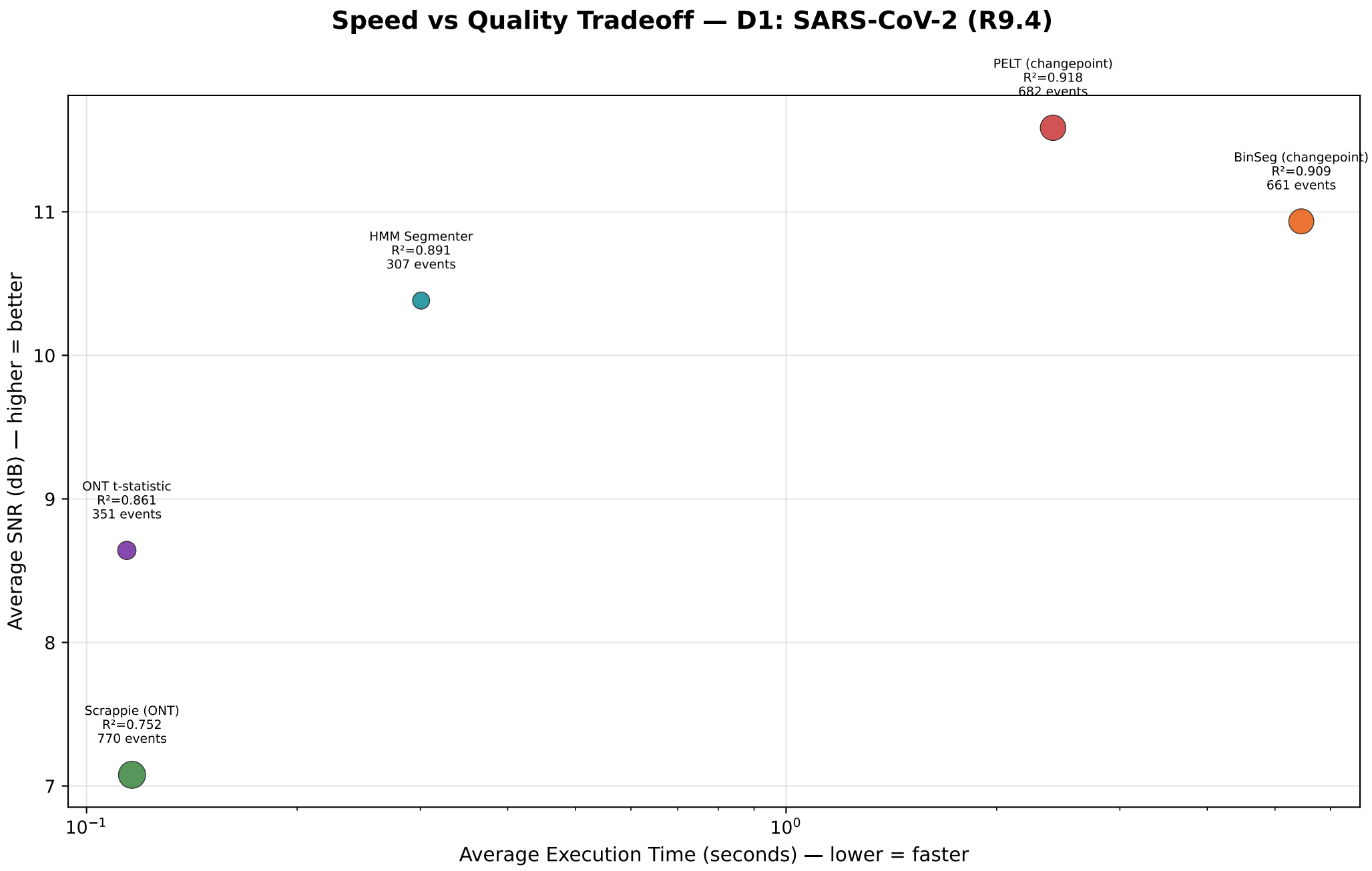


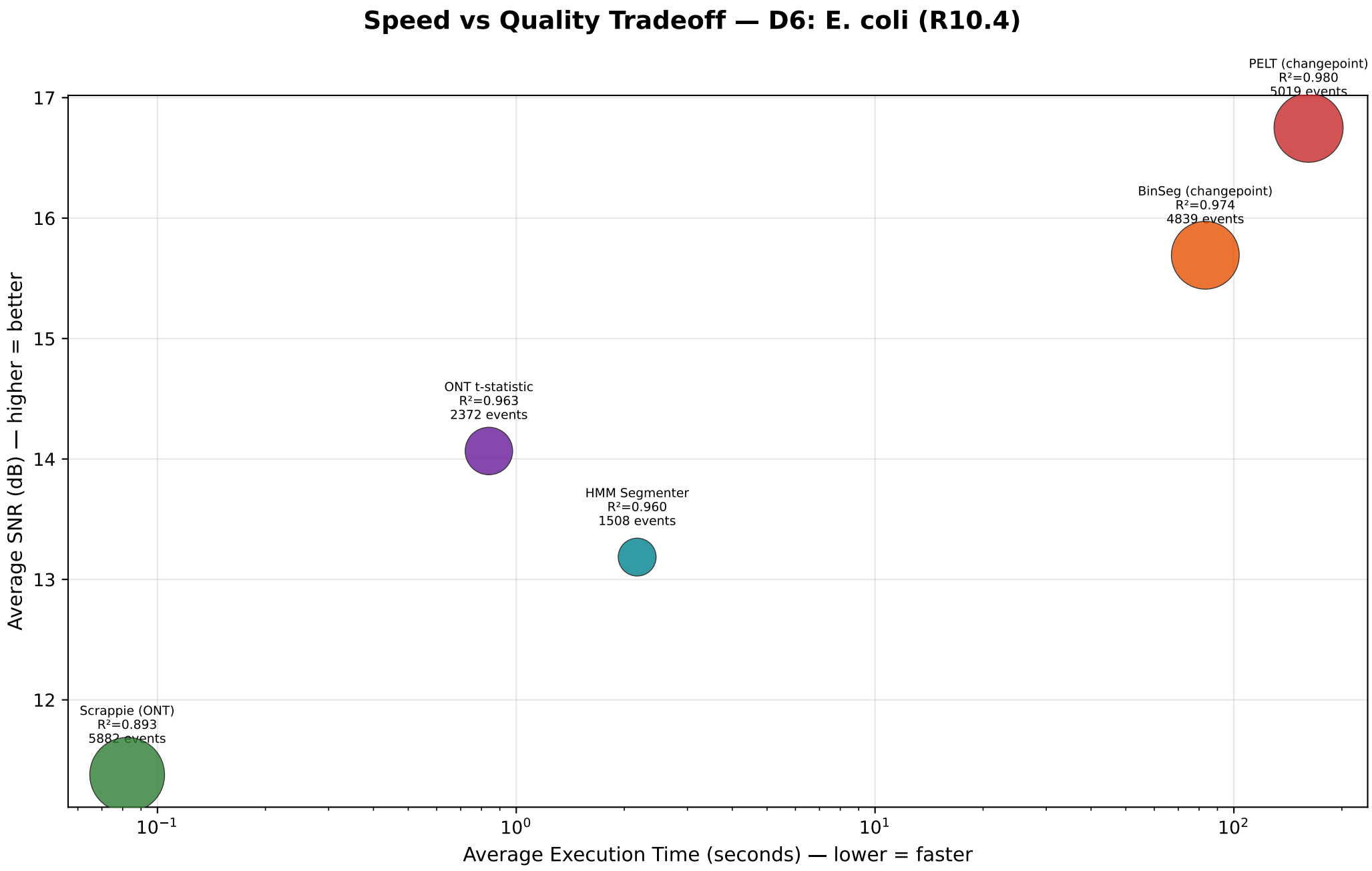


Segmentation Consistency Across Reads — D6: E. coli (R10.4)






Speed vs Quality Tradeoff — D1: SARS-CoV-2 (R9.4)
PELT (changepoint)
R²=0.918
682 events
BinSeg (changepoint)
R²=0.909
661 events
HMM Segmenter
R²=0.891
307 events
ONT t-statistic
R²=0.861
351 events
Scrappie (ONT)
R²=0.752
770 events
Average SNR (dB) — higher = better
Average Execution Time (seconds) — lower = faster
11
10
9
8
7
10^{-1}
10^{0}


Speed vs Quality Tradeoff — D6: E. coli (R10.4)
PELT (changepoint)
R²=0.980
5019 events
BinSeg (changepoint)
R²=0.974
4839 events
ONT t-statistic
R²=0.963
2372 events
HMM Segmenter
R²=0.960
1508 events
Scrappie (ONT)
R²=0.893
5882 events
Average SNR (dB) — higher = better
Average Execution Time (seconds) — lower = faster
17
16
15
14
13
12
10^{-1}
10^{0}
10^{1}
10^{2}

Cross-Dataset Comparison

Algorithm	Category	SNR (D1: SARS-CoV)	R^2 (D1: SARS-CoV)	SNR (D6: E. coli)	R^2 (D6: E. coli)
Scrappie (ONT)	A: Nanopore tool	7.08	0.7515	11.38	0.8934
ONT t-statistic	B: Nanopore algorithm	8.64	0.8614	14.07	0.9625
HMM Segmenter	B: Nanopore algorithm	10.38	0.8911	13.19	0.9599
PELT (changepoint)	C: General baseline	11.59	0.9179	16.75	0.9797
BinSeg (changepoint)	C: General baseline	10.93	0.9089	15.69	0.9743